

Gauging- δ : A Non-Parametric Hierarchical Clustering Algorithm

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Abstract—The development of a nonparametric and versatile clustering algorithm has been a longstanding challenge in unsupervised learning due to the exploratory nature of the clustering problem. This study presents a novel algorithm, named Gauging- δ , which can handle diverse cluster shapes and operate in a non-parametric manner. The algorithm employs a hierarchical merging process that starts from individual data points until no further clusters can be merged. The central component of Gauging- δ is the adaptive mergeability function, which progressively determines if two clusters are mergeable considering the perceptual statistics of the clusters and their environment. Empirical evaluations on 105 synthetic datasets demonstrate the superiority of the proposed algorithm, particularly in accurately handling well-separated clusters. Experiments on real-world datasets highlight the impact of selecting appropriate data features and distance metrics on clustering results.

Index Terms—Hierarchical clustering, non-parametric algorithm, cluster shapes.

I. INTRODUCTION

CLUSTERING algorithms play a pivotal role in the analysis and interpretation of complex datasets across various scientific disciplines, enabling the discovery of inherent structures and patterns by grouping similar entities together [1], [2]. Traditional clustering techniques, such as K-Means [3], have demonstrated significant success in applications ranging from image analysis to genomics [4].

However, these methods often rely on predefined assumptions about the data's geometry, typically favoring spherical or convex cluster shapes, and require the specification of the number of clusters in advance [5], [6]. These intrinsic limitations pose substantial challenges in real-world scenarios, where the underlying data distribution may not conform to simplistic geometric constraints [7], [8] or where determining the correct number of clusters is not straightforward. The development of a

non-parametric algorithm that can adaptively handle a variety of cluster shapes without requiring prior knowledge of the number of clusters is therefore of paramount importance.

Recent advancements in non-parametric clustering algorithms have introduced innovative approaches to overcoming traditional challenges in the field [9], [10]. Mukhoty et al. [11] introduced a parameter-free, affinity-based method that excels in simplicity and robustness against noise and density variations, albeit with limitations in handling non-convex data shapes. UNIC (UNparameterized Clustering) [12], focusing on large datasets, improves on time and space complexity and reduces the need for control parameter adjustments, yet struggles with clustering quality and stability. Rehman and Belhaoui's divide-and-merge algorithm [13] showcases versatility in identifying clusters of various shapes and densities and in outlier detection without requiring a predefined cluster count, though it requires fine-tuning of several parameters. The border-peeling clustering [14] method effectively separates adjacent clusters by adapting to local densities but faces difficulties with non-uniform distributions and relies on critical parameter selection [15]. These developments underscore the ongoing quest for algorithms that can adaptively navigate the complexities of real-world data, especially concerning non-convex shapes and parameter dependency.

Both spectral clustering and density-based clustering are well-suited for identifying complex cluster shapes that go beyond simple geometric assumptions. Spectral clustering excels at capturing non-linear separations in data, but it requires the number of clusters to be specified a priori. Moreover, its performance is sensitive to the choice of similarity measures and the construction of the affinity matrix [16], [17], which can complicate parameter selection [18]. Similarly, density-based algorithms, such as Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [19] and Density Peak (DP) clustering [20], are highly effective at detecting clusters of arbitrary shapes and sizes. However, their performance is also strongly influenced by parameter choices, particularly with regard to the density thresholds [21].

Hierarchical clustering is a flexible method that can utilize different linkage measures, such as single-linkage, complete-linkage, or average-linkage, to build a tree-like structure of nested clusters. The single-linkage hierarchical clustering algorithm [22] is claimed to be effective in various data shapes yet suffers from the chaining effect, i.e., clusters connected by a line of few close points cannot be separated [23]. What's more, hierarchical clustering algorithms require either cluster number

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or distance threshold to decide the stopping point when obtaining the tree-like cluster structure [24].

In eliminating the recursive dependence of a clustering algorithm on the clustering results, we have proposed an automatic hierarchical clustering algorithm in this paper, which is parameter-free and independent of the properties of the data set to be clustered. The proposed algorithm Gauging- δ was developed by constructing an adaptive mergeability function by gauging the $d(\delta)$ distance between clusters. Same idea has been successfully applied in the construction of a parameter-free algorithm for reconstructing curves from samples of various shapes and properties [25], [26].

For this paper, the main contributions are as follows:

- 1) Introduction of Gauging- δ , a non-parametric hierarchical clustering algorithm that effectively handles diverse data shapes. The proposed algorithm is guided by an adaptive mergeability function leveraging cluster perceptual statistics and environmental context.
- 2) Proposed two perceptual statistics: proximity and continuity to decide the mergeability of two arbitrary clusters, without the need to specify the number of clusters.
- 3) Extensive evaluations on 105 synthetic datasets demonstrate the method's superiority in accurately identifying well-separated clusters, and additional experiments on real-world datasets underscore the importance of selecting suitable data features and distance metrics.

The structure of this paper is as follows: Section II details the clustering process and explicates the algorithm proposed in this study. In Section III, we present the experimental results of our algorithm and compare its performance against several well-known clustering algorithms. The effects of the mergeability function on the clustering outcomes are discussed in Section IV. Section V offers concluding remarks and summarizes the paper.

II. THE PROPOSED ALGORITHM

A. Cluster Generation Process

The proposed algorithm is designed to progressively merge clusters based on their mergeability, which assumes that every data sample is a cluster. Two nearest clusters C_i and C_j can be merged if and only if they meet the mergeability criteria. The input of the algorithm is a given dataset $\mathcal{X} = \{\mathbf{x}_i | i = 1, \dots, n\}$, where $\mathbf{x}_i \in \mathbb{R}^d$, whereas the output is a set of clusters $\mathcal{C} = \{C_i | i = 1, \dots, k; k \leq n\}$. The value of k equals n at the beginning of the algorithm but decreases along with the merging of clusters.

First, a distance matrix $\mathcal{D}$ is constructed for each pair of clusters, where the cluster distance d_{ij} between clusters C_i and C_j is the euclidean distance between the two nearest points, $r_i(\in C_i)$ and $r_j(\in C_j)$. These two points are named reference points throughout this article. An ascending distance list D' is then obtained from the matrix $\mathcal{D}$.

Second, the first pair of clusters C_1 and C_2 is selected from the list D' to calculate the mergeability. If they are mergeable, a new cluster is generated by merging them as $C_i = C_1 \cup C_2$; all the cluster distance pairs containing C_1 and C_2 are removed

Algorithm 1: Gauging- δ Clustering Process.

Input: n initial clusters $C_1, \dots, C_n$, with each cluster being a single data point $\mathbf{x}_i \in \mathbb{R}^d$.
Output: k merged clusters $C_1, \dots, C_k$.
Initialize: Distance matrix D , where d_{ij} is the distance between any two clusters C_i and C_j .
1 Sort D to get an ascending distance list D' ;
2 **while** $D' \neq \emptyset$ **do**
3 Set $M \leftarrow \text{currentClusterCount}$;
4 **foreach** d'_{ij} in D' **do**
5 Identify corresponding clusters C_i and C_j ;
6 Check mergeability of C_i and C_j using Mergeability Function;
7 **if** mergeable **then**
8 Merge C_i and C_j ;
9 $M \leftarrow M - 1$;
10 Record newly generated cluster and remove merged clusters;
11 **end**
12 **end**
13 **if** M equals $\text{currentClusterCount}$ **then**
14 **break** // No new cluster generated
15 **end**
16 **else**
17 Update D' for the new list of clusters;
18 **end**
19 **end**

from D' . Otherwise, the next pair of clusters is selected to repeat the above mergeability check until no pair is in the list. A new list of clusters is comprised of non-mergeable clusters and newly generated clusters.

Last, the list D' is updated and re-ordered for this new list of clusters. The above step will be repeated until no single pair of clusters is merged. The whole clustering process, named Gauging- δ , is shown in Algorithm 1.

The core of the Gauging- δ algorithm is the definition of mergeability function. Please note that the mergeability function does not depend on the shape of the data; instead, it only depends on the nature of data from the concerned domain. We argue that two clusters are mergeable if they are close enough and continuous. In the following, we define the mergeability function accordingly.

B. Adaptive Mergeability Function: Proximity

The proximity function is designed by following two intuitive observations (Fig. 1): (1) If the data samples in the first cluster are sparse and not uniform, the chances of the first cluster taking in the second one will increase; (2) Two clusters are more likely to merge if they are far from other clusters in their environment. The environment E_i of Cluster C_i consists of all existing clusters, excluding C_i . The two factors α_i and β_{ij} in (3) reflect the two proximity principles, respectively.

In the clustering process, each cluster accumulates the distances at which merges occur, known as its *historical merging*

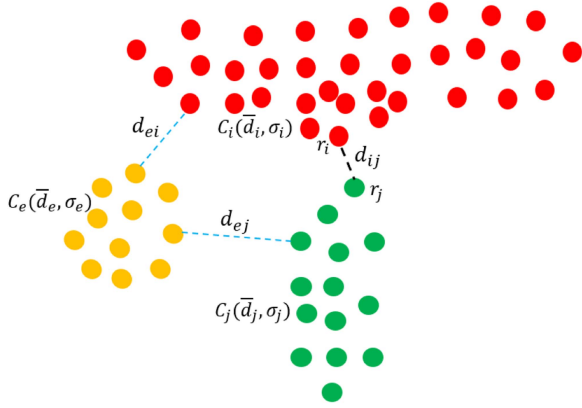


Fig. 1. Factors influencing proximity. The proximity is determined by the properties of clusters C_i , C_j and the environment cluster C_e . $\bar{d}_i(\bar{d}_j, \bar{d}_e)$ is the mean of the historical merging distance of cluster $C_i(C_j, C_e)$. $\sigma_i(\sigma_j, \sigma_e)$ is the standard deviation of the historical merging distance of cluster $C_i(C_j, C_e)$. $d_{ij}(d_{ei}, d_{ej})$ is distance between cluster C_i and C_j (C_i and C_e , C_j and C_e). Two reference points are marked as r_i and r_j .

distances. Initially, each data point is treated as a singleton cluster with no merging distance. Whenever two clusters C_i and C_j merge at distance d_{ij} , the newly formed cluster inherits the historical distances from both clusters plus the new distance d_{ij} . As the algorithm iterates, each cluster thus compiles a growing list $\{d_1, d_2, \dots, d_{m_i}\}$ of these merge distances. The average historical merging distance $\bar{d}_i$ for a cluster C_i is the arithmetic mean of all distances in that list:

$$\bar{d}_i = \frac{1}{m_i} \sum_{k=1}^{m_i} d_k, \quad (1)$$

and the standard deviation σ_i is computed as

$$\sigma_i = \sqrt{\frac{1}{m_i - 1} \sum_{k=1}^{m_i} (d_k - \bar{d}_i)^2}. \quad (2)$$

When C_i and C_j are both singletons, they don't have merging history, $\bar{d}_i(\bar{d}_j)$ and $\sigma_i(\sigma_j)$ do not exist. In this case, we consider only global requirements. They pass proximity if $d_{ij} \leq 1.5 * d_{min}$, where d_{min} is the minimum distance among all the clusters in the current state. Otherwise, the mergeability of C_i and C_j is decided by the mergeability function (3).

There are two scenarios as follows:

- 1) *Point (C_i) - Cluster (C_j):* Do single side judgment, i.e. as long as $d_{ij} \leq d_j^*$, they pass the proximity test; and
- 2) *Cluster-Cluster:* If $d_{ij} \leq d_i^*$ and $d_{ij} \leq d_j^*$, they pass the proximity test.

$$d_i^* = \bar{d}_i \cdot \alpha_i \cdot \beta_{ij}. \quad (3)$$

The average historical merging distance $\bar{d}_i$ is the base of a cluster's expansion, which provides a local metric base to measure the proximity of two candidate clusters.

$$\alpha_i = \frac{4}{1 + e^{-4 \frac{\sigma_i}{\bar{d}_i}}} - 1. \quad (4)$$

The merging-distance factor α_i (4) defines the expansion degree based on the distribution of samples in the cluster C_i . If the data samples in the cluster are sparse and unevenly distributed,

the likelihood of the cluster merging with the other one increases. In contrast, if the cluster tends to have uniform distribution, merging with a new cluster becomes difficult because it may break the uniformity. In the extreme case of uniform distribution ($\sigma_i = 0$), α_i becomes 1, making it the most difficult to merge two clusters. The uniformity of a cluster is measured by the normalized standard deviation of its historical merging distance.

When a cluster has two points, $\sigma = 0$, $\alpha_i = 1$, no new cluster can be merged in, which often happens in the beginning stage of the cluster merging process. In this case, we set $\alpha_i = 2$, which works well in empirical experiments.

The environment scale β_{ij} (5) takes into account the impact of an environment cluster $C_e (\in E_{ij})$ on the mergeability of clusters C_i and C_j . As defined above, the environment E_{ij} of Cluster C_i and C_j consists of all existing clusters except C_i and C_j . The further C_e is to both C_i and C_j , the easier C_i and C_j can be merged.

$$\beta_{ij} = \frac{2}{1 + e^{5\omega}} + 1, \quad (5)$$

$$\omega = \frac{d_{ij}}{(d_{ie} + d_{je})/2}, \quad (6)$$

where d_{ie} (d_{je}) is the distance between cluster C_i (C_j) and the most influential environmental cluster C_e among E_{ij} .

C_e is the cluster that is near to both C_i and C_j and has a comparatively large size. C_e is the cluster of the highest F value (7) among E_{ij} . The equation modifies Newton's law of universal gravitation to illustrate that the impact of an environment cluster on the clusters under consideration is determined by two factors: the number of samples within the environment cluster and the distance between the environment cluster and the considered cluster. It must be noted that a similar idea was independently proposed in the Torque Clustering algorithm [27] to measure clusters' connection.

$$F = \sqrt{\frac{|C_i| |C_e| |C_j| |C_e|}{d_{ie}^2 d_{je}^2}}, \quad (7)$$

where $|C_*|$ is the number of points in cluster C_* .

Obviously, α_i (α_j), β_{ij} and $\bar{d}$ are all adaptive to the local properties of the given data set D and do not depend on any unknown properties of the clusters to be produced.

Once the two clusters meet the proximity criterion, the algorithm will proceed to check continuity; otherwise, they cannot be merged. Two clusters can be merged only if they pass both proximity and continuity tests.

C. Adaptive Mergeability Function: Continuity

Continuity function can be defined by using cross-cluster discontinuities, which occur when there are abrupt changes in shape, density, and distribution in transitional areas crossing the two clusters. The discontinuity is measured in the local transitional area (Fig. 2). Specifically, the following continuation metric, ξ_{ij} , is used to measure the degree of continuation between two clusters, C_i and C_j .

$$\xi_{ij} = \xi_d \cdot \sqrt{\xi_a \cdot \xi_o \cdot \xi_s}. \quad (8)$$

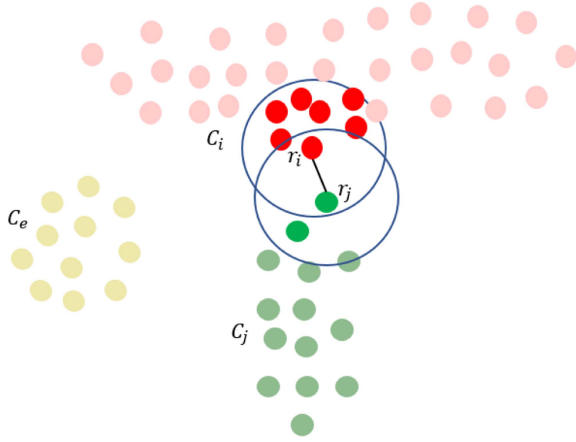


Fig. 2. Adaptive continuation function. The two circles marked the local area in which the circled points (dark color) decide the continuation value of cluster C_i and C_j .

In this paper, a constant continuation threshold ξ_{ij}^* , set to 0.15, is used to determine if two clusters have good continuation. If $\xi_{ij} > \xi_{ij}^*$, two clusters pass the continuity test. The threshold, ξ_{ij}^* , independent of the data set to be clustered, can be further optimized based on the nature and experiments for different application domains.

A lower density continuation, ξ_d , is expected if a decrease in the number of points is detected when moving from far to near the reference points. Angle transition ξ_a is a measurement of the smoothness of two clusters' transition. The closer the boundary angle to $\pi/2$, the sharper the transition is. Orientation similarity ξ_o is calculated by comparing the similarity of the mass between the two clusters. Shape similarity ξ_s measures the similarity of the two clusters' shapes, captured by the variance of the distance between points and the cluster center in the clustering growing process. The calculation of local transitional area, density continuation, angle transition, orientation similarity and shape similarity is demonstrated in the following.

1) *Local Transitional Area*: In this paper, the local transitional area is defined by a circular range with center r_i or r_j and a radius R . Several radiuses are tested from small to large, it is believed the continuation is good as long as two clusters pass one test. Specifically, the radius is set as:

$$R = \delta \cdot d_{base}, \quad (9)$$

where δ takes value from $\{2, 2.5, 3\}$.

$$\begin{aligned} d_{base} &= \frac{d_{ij} + \max(D_{latest-5})}{2} \cdot \frac{2}{1 + e^{(-\frac{\varphi}{20})}} \\ &= \frac{d_{ij} + \max(D_{latest-5})}{1 + e^{(-\frac{\varphi}{20})}}, \end{aligned} \quad (10)$$

where $D_{latest-5}$ is the latest five merging distances of cluster C_i 's historical merging distances $\{d_1, d_2, \dots, d_{m_i}\}$.

$$\varphi = \max\left(\frac{d_{ij}^c}{d_{ij}}, 3\right) - 3, \quad (11)$$

where d_{ij}^c is the distance between the centroid of the two clusters C_i and C_j . The centroid c_i of a cluster C_i is the arithmetic mean

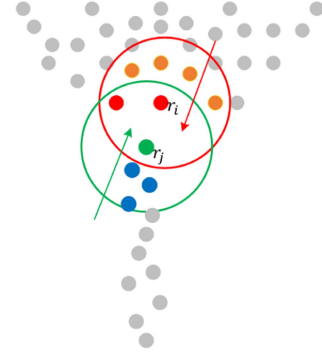


Fig. 3. Density continuation ξ_d . The points in the red circle are P_i , and those in the green circle are P_j . The orange points are P_{i-j} , and the blue ones are P_{j-i} . P_{ij}^i are the red points in the intersection area; P_{ij}^j are the green points in the intersection area. The density is decreasing along the direction of the arrows.

of all the data points in the cluster. Let cluster C_i contain n_i data points $\{x_1, \dots, x_{n_i}\}$. c_i is calculated as:

$$c_i = \frac{1}{n_i} \sum_{k=1}^{n_i} x_k. \quad (12)$$

τ is a weighted measurement of the two clusters' compactness.

$$\tau = \frac{1}{|C_i| + |C_j|} \sum_{l=i,j} |C_l| \frac{d_{m_i}}{d_l}. \quad (13)$$

Once the local transitional area is decided, points within the range of r_i and r_j constitute set P_i and P_j , respectively.

2) *Density Continuation ξ_d* : ξ_d is evaluated by counting the number of points in the divided four sets of P_i and P_j (Fig. 3). $P_{i-j}(P_{j-i})$ constitutes of the samples in $P_i(P_j)$ but not in $P_j(P_i)$. $P_{ij}^i(P_{ij}^j)$ constitutes of the common points of P_i and P_j that are closer to $r_i(r_j)$. $n_i(n_j)$, $n_{i-j}(n_{j-i})$ and $n_{ij}^i(n_{ij}^j)$ are the cardinality of the corresponding sets. ξ_d is calculated by (14), if n_{ij}^i or n_{ij}^j is the minimum value among n_{i-j} , n_{j-i} , n_{ij}^i and n_{ij}^j . Otherwise, $\xi_d = 1$.

$$\xi_d = \frac{\min(n_{ij}^i, n_{ij}^j)}{\max(n_{i-j}, n_{j-i})}. \quad (14)$$

3) *Angle Transition ξ_a and Orientation Similarity ξ_o* : r_M is a generated middle point from the two reference points $r_i r_j$. For every two points p_{ik} and p_{il} in P_i , $\angle p_{ik} r_M p_{il}$ is called rotation angle. The two biggest rotation angles θ_i and θ_j are found from the two clusters, respectively. The corresponding four boundary points p_{ik}^* , p_{il}^* , p_{jk}^* and p_{jl}^* are identified.

We then find the smallest angle from the four angles $\angle p_{ik}^* r_M p_{jk}^*$, $\angle p_{ik}^* r_M p_{jl}^*$, $\angle p_{il}^* r_M p_{jk}^*$ and $\angle p_{il}^* r_M p_{jl}^*$ formed by the four boundary points, denoted as θ_L . The remaining two points form angle θ_R . θ_L and θ_R are boundary angles (Fig. 4). ξ_a is calculated as (15).

Orientation similarity ξ_o (16) is defined as the ratio of the density, calculated as the ratio of the number of points in the

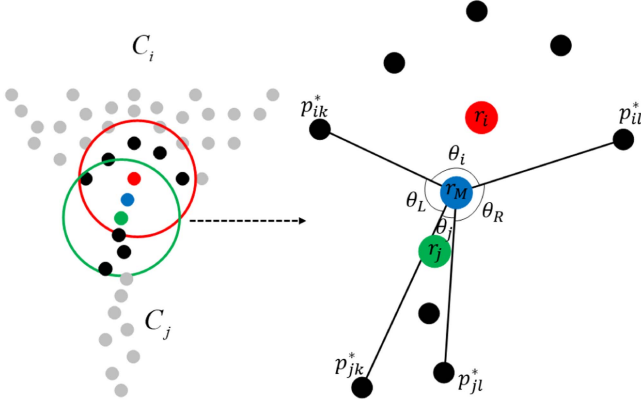


Fig. 4. Angle transition ξ_a and orientation similarity ξ_o . The points (right side) in the figure are extracted from the local area (left side). The red and green points are the two reference points from the two clusters, respectively. The blue point r_M is the generated middle point from the two reference points. The angles formed by r_M and the surrounding points are used to calculate angle transition and orientation similarity.

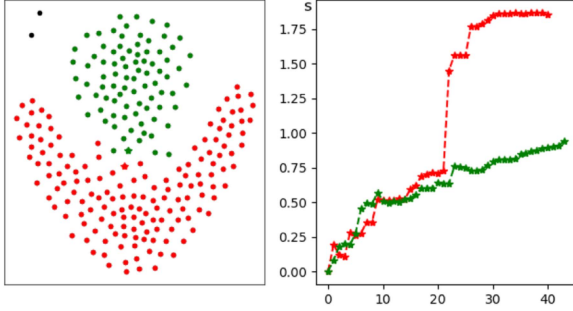


Fig. 5. Historical shape of the two clusters. The right figure shows the change in the shapes of the two clusters (red and green in the left figure) after each merge.

local area over the area of the local area.

$$\xi_a = \frac{\left(\frac{|\theta_L - \pi/2|}{\pi/2} + \frac{|\theta_R - \pi/2|}{\pi/2}\right)}{2} = \frac{|\theta_L - \pi/2| + |\theta_R - \pi/2|}{\pi}. \quad (15)$$

$$\xi_o = \frac{n_i/A_i}{n_j/A_j} = \frac{n_i/\theta_i R^2}{n_j/\theta_j R^2} = \frac{n_i \cdot \theta_j}{n_j \cdot \theta_i}. \quad (16)$$

4) *Shape Similarity ξ_s* : The shape similarity ξ_s quantifies the similarity between the shapes of two clusters and is calculated using (17).

$$\xi_s = \frac{\phi}{1 + \phi} + 0.5, \quad (17)$$

$$\phi = \frac{\min(\sigma_s^i, \sigma_s^j)}{\max(\sigma_s^i, \sigma_s^j)}, \quad (18)$$

where σ_s is the standard deviation of a cluster's historical shape. The "historical shape" of a cluster records how its shape metric s_i evolves at each stage of the iterative merging process. As an example, Fig. 5 displays the changing of the shape of the two clusters in the "flame" dataset over time.

The shape s_i of a cluster C_i is calculated as the standard deviation of the distance between the samples in a cluster and

the cluster's centroid c_i . s_i serves as a measure that distinguishes clusters based on their shape, with larger values for more elongated or irregular distributions (like stripes) and smaller values for compact, symmetric shapes (like circles).

$$s_i = \sqrt{\frac{1}{n_i} \sum_{k=1}^{n_i} (\|x_k - c_i\| - \mu_i)^2}, \quad (19)$$

where μ_i is the mean distance between samples in cluster C_i and its centroid, given by:

$$\mu_i = \frac{1}{n_i} \sum_{k=1}^{n_i} \|x_k - c_i\|. \quad (20)$$

Again, it must be noted that the mergeability function is independent of the properties of the data sample. The application of the algorithm will rely on the nature of the problems rather than the data itself. Experiments in different areas can be done to decide the corresponding equations, which makes the proposed algorithm parameter-free.

D. Time Complexity Analysis

The overall procedure of the algorithm is similar to agglomerative hierarchical clustering, whose time complexity is $O(n^2 \log n)$ or $O(n^3)$, depending on the data structure used. We assume the optimal complexity $O(n^2 \log n)$ can be achieved by keeping track of the nearest clusters using advanced data structures. The difference between our algorithm and the hierarchical clustering algorithm is the extra mergeability assessment. The calculation of the mergeability function is considered at a constant level.

If each time the nearest pair of clusters can pass the mergeability test, the time complexity will be the same as the hierarchical clustering $O(n^2 \log n)$. The smallest complexity will happen if we check until the last pair and they cannot merge. In this case, the clustering process stops in one iteration and the complexity is $O(n^2)$. The biggest complexity $O(n^3)$ will happen if each iteration, the furthest pair of clusters merge. In summary, the time complexity ranges from $O(n^2)$ to $O(n^3)$ depending on the mergeability of clusters.

III. EXPERIMENTS

A. Data Sets

To assess the performance of the proposed clustering algorithm, both 2-D and 3-D synthetic data that are convenient for visualization and high dimensional real-world data sets are included in the experiments. All the data sets' ground truth labels are known.

Ten benchmark data sets have been selected to demonstrate the proposed algorithm's performance based on clusters' properties and spatial relations. Their characteristics and targeted challenges are displayed in Table I. Data sets '3blobs', 'spiral', and 'Jain' are all well-separated but have different shapes. The data set 'compound' presents several challenges. Data sets 'flame', 'aggregation' and 'pathbased' are not well-separated in different forms. The data set 'impossible' has been selected mainly for testing the algorithm's ability to handle background

TABLE I
DATA SETS AND THEIR CHARACTERISTICS

Name	Targeted problem	Dimension	No. of samples	No. of clusters	Source
3blobs	baseline	2	600	3	[28]
spiral	non-linearly separable, nested	2	312	3	[29]
Jain	non-convex, non-linearly separable, uneven density	2	373	2	[30]
compound	noise, nested clusters	2	399	6	[31]
flame	non-linear separation, overlapping, outlier	2	240	2	[32]
aggregation	diverse shapes, touching contour, non-convex, uneven-sized clusters	2	788	7	[33]
pathbased	Gaussian distribution, overlapping, nested	2	300	3	[29]
impossible	background noise, compound	2	3673	7	[6]
chainlink	non-linearly separated	3	1000	2	[34]
atom	non-linearly separated	3	800	2	[34]

noise. Data sets ‘chainlink’ and ‘atom’ are of three dimensions and non-linearly separable.

To further verify the superiority of the proposed algorithm, the algorithms are tested and contrasted on 95 more 2-D and 3-D (see Table S2 in supplementary material) synthetic data sets [35] besides the benchmark data sets. The 105 data sets in total are classified into two categories: well-separated and not well-separated [36]. Well-separated data sets refer to the case that any two clusters are well-separated between each other. These clusters can be either convex or non-convex, as long as they are sufficiently separated. Conversely, data sets where clusters are not well-separated fall into not well-separated category. These can be further classified into three types: thin bridge connection, large area touching, and noise.

The real-world experiments are conducted on the widely recognized USPS dataset, which is used for handwritten character recognition, to further evaluate the algorithm’s robustness in a real-world visual context. The USPS dataset consists of ten clusters representing the digits 0-9, with 7291 training samples and 2007 testing samples. Each sample is a 16×16 grayscale image, resulting in 256 features. Clustering is performed on the testing examples, while the training examples are used to train a convolutional neural network (CNN) to obtain the feature embeddings.

B. Algorithms for Comparison

Seven other representative algorithms are selected for comparison with the proposed algorithm, that is, K-Means++ [37], GMM (Gaussian Mixture Models) [38], HDBSCAN (Hierarchical DBSCAN) [39], SingleLinkage [22], MST (Minimal Spanning Tree) [31], spectral clustering [40] and BP (Border-Peeling) [14], which is partition-based, distribution-based, density-based, hierarchical, graph-based, spectral and non-parametric, respectively. K-Means++, GMM, SingleLinkage, and spectral clustering require the provision of ground-truth cluster numbers as input parameters. In contrast, the parameters for HDBSCAN and MST are consistently maintained across all datasets, hence viewed as non-parametric here. Additionally, we have included a comparative analysis with the BP algorithm, a non-parametric clustering method recently introduced by Averbuch-Elor et al. [13], which was reported to have the best performance among many non-parametric clustering algorithms.

The parameter setting in the algorithms is as follows. K (the number of target clusters) in K-Means++, GMM, SingleLinkage, and spectral clustering is set as the ground truth number of clusters. The cut-off scale in MST is set to 2. The affinity matrix in spectral clustering is constructed using a radial basis function (RBF) kernel.

A variant of Gauging- δ , denoted as Gauging- δ (proximity), is included for comparison. In this variant, the continuation threshold ξ_{ij}^* (8) is set to zero, allowing two clusters to merge as long as they pass the proximity test.

C. Evaluation

This section presents a comprehensive evaluation of Gauging- δ under various conditions. First, we evaluate its effectiveness on a range of synthetic benchmark datasets, providing a verification of its capability on various cluster shapes. We then analyze how the algorithm handles both well-separated and not well-separated clusters, showcasing its advantages in dealing with well-separated data. Next, we compare its behavior against both parametric and non-parametric methods to highlight key distinctions and benefits. Finally, we assess its applicability on real-world datasets to further validate its robustness and practical value.

1) *Verification on Synthetic Benchmark Data Sets:* The clustering results generated by Gauging- δ on the ten benchmark data sets are displayed in Fig. 6. In contrast, those from Gauging- δ (proximity) are given in Table S1 in the supplementary materials.

The experiments are first done on the ‘3blobs’ data set, where three convex clusters are well-separated. Every clustering algorithm is expected to perform correctly on this baseline problem. Gauging- δ separates the three blobs almost correctly, except one border point is identified as an outlier.

The algorithm does not have difficulty in recognizing the three nested spirals in ‘spiral’, though one spiral is broken near the tail. Gauging- δ successfully separates the two half-rings of the ‘Jain’ data set [30] that have unequal density. The ‘chainlink’ data set consists of intricate links of two chains intertwined in R^3 . The center of each ring is closer to some elements of the other cluster than to elements of its own cluster, which is difficult for algorithms like K-Means++. Gauging- δ can accurately distinguish the two chains. The outer hull and the inner core of the ‘atom’ data set are identified correctly.

The background noise (right part) and nested clusters (lower left) challenges are resolved well in the ‘compound’ data set,

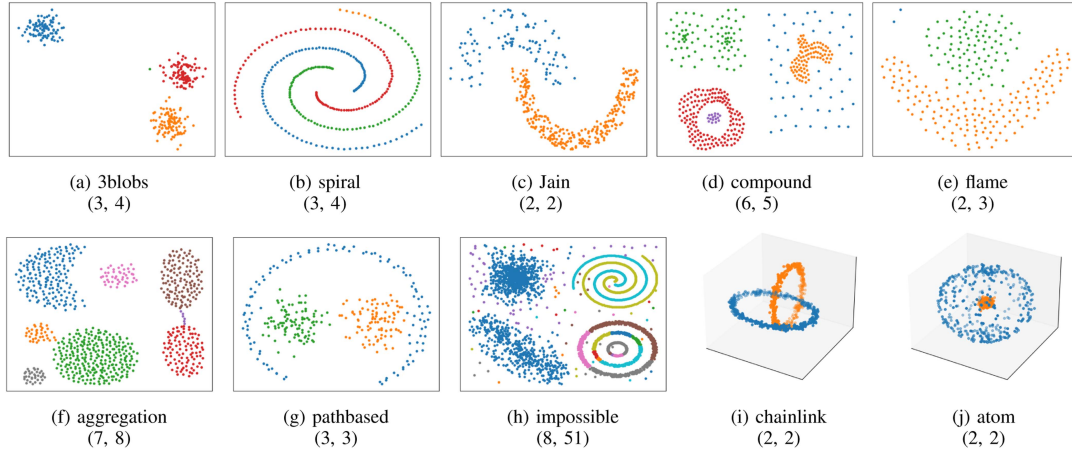


Fig. 6. Experiments results on ten representative benchmark data sets by Gauging- δ . The pair of numbers under each subfigure indicates (the ground-truth cluster numbers, the detected cluster numbers).

while the two symmetrical clusters (left up) are not separated. It is reported on the ‘impossible’ data set that “Although these clusters are apparent to a data analyst, none of the available clustering algorithms can detect all these clusters.” [6]. Generally, Gauging- δ has detected the seven clusters despite the existing background noise. Though the three right lower corner circles are divided finely into small pieces, it is an acceptable under-clustering, which maintains the structure of the cluster, compared with grouping samples into one cluster that destroys the cluster structure.

The three not well-separated data sets (‘flame’, ‘aggregation’ and ‘pathbased’) are solved by considering the shape, orientation, and density factors, respectively. Three clusters are correctly identified in the ‘flame’ data set, including the two outlier points, consistent with the original paper’s result [32]. For the ‘aggregation’ problem, no single method can correctly detect all seven clusters defined by the original paper [33]. The touching clusters with a narrow bridge between them pose challenges to algorithms assuming close data should be clustered together. Gauging- δ has determined eight clusters, whereas seven clusters are usually believed, since Gauging- δ considers that the thin line connecting the two oval clusters should be an individual cluster, which is more intuitive to human perception. For the ‘pathbased’ data set [29], Gauging- δ can distinguish the circular cluster and the two blobs inside.

Gauging- δ does not require cluster number as a parameter designated to data sets and can stop the merging process automatically if no pair of clusters meets the proximity and continuity criteria. Out of the ten benchmark data sets, Gauging- δ has detected the same number of clusters with the ground truth for four data sets (‘Jain’, ‘pathbased’, ‘chainlink’, and ‘atom’). For the remaining data sets, our algorithm identifies more clusters (‘flame’, ‘aggregation’, ‘3blobs’, and ‘spiral’) but the additional clusters could be considered as outliers or independent clusters (e.g., the line between the two oval shapes in ‘aggregation’). In the ‘impossible’ example (h), our algorithm does not group background noise into a single cluster, which results in substantial clusters with most of them having only one point than the ground truth. These extra clusters could be easily detected as outliers or background using post-processing methods.

Therefore, our proposed algorithm is able to cluster datasets of different shapes, densities, and distributions. Both Gauging- δ and Gauging- δ (proximity), clearly outperform other algorithms on the ten benchmark problems, as shown in Tables II and III.

2) Results Analysis on Well-Separated and Not Well-Separated Data: For 53 well-separated data sets, Gauging- δ (proximity) achieves the highest average NMI and ARI (both 0.94), along with the lowest standard deviations (less than 0.08), among all the comparison algorithms, as shown in Tables IV and V. Meanwhile, for 52 not well-separated data sets, Gauging- δ (proximity) performs better only than SingleLinkage and MST, which are also distance-based methods. Furthermore, Gauging- δ (proximity) can handle many not well-separated data sets, achieving high performance on 19 out of 52 data sets, with both NMI and ARI exceeding 0.8 (see Tables S4 and S6 in supplementary materials). This capacity is attributed to (1)–(4), which inherently account for certain density features of the data. This is further confirmed by the benchmark problems shown in Fig. S1.

In an effort to improve the algorithm performance of Gauging- δ (proximity) in handling not well-separated data sets, Gauging- δ incorporates the continuity component into its mergeability function. As a result, the algorithm demonstrated higher performance on ten benchmark data sets, with ‘flame’, ‘aggregation’ and ‘pathbased’ serving as representatives of not well-separated clusters, as shown in Fig. 6. The main improvement lies in the handling of data sets with bridge connections. Compared to Gauging- δ (proximity), Gauging- δ improved the performance in clustering 52 not well-separated data sets, with 23 of these achieving both NMI and ARI scores above 0.8 (see Tables S4 and S6 in the supplementary materials). However, its overall performance decreased because of under-merging caused by overly stringent continuity conditions. This indicates that while continuity remains a challenge requiring further investigation, there is a potential to define domain specific continuity functions that can cluster data sets effectively, even without prior knowledge of their structure.

Therefore, Gauging- δ (proximity) is well-suited for well-separated data sets and can effectively handle many not

TABLE II
NMI OF THE TEN BENCHMARK PROBLEMS FROM THE EIGHT ALGORITHMS

Data set	K-Means++	GMM	SingleLinkage	HDBSCAN	MST	BP	Spectral	Gauging- δ	Gauging- δ (proximity)
3_blobs	0.983	0.983	0.983	0.938	0.734	0.951	0.983	0.975	0.975
spiral	0.001	0.001	1	0.284	1	0.205	1	0.968	1
Jain	0.369	0.202	0.246	0.157	0.729	0.451	1	1	1
compound	0.714	0.768	0.799	0.776	0.761	0.791	0.767	0.952	0.952
flame	0.394	0.345	0.024	0.424	0.024	0.908	0.024	0.971	0.971
aggregation	0.842	0.912	0.883	0.873	0.889	0.988	0.982	0.981	0.873
pathbased	0.548	0.525	0.013	0.752	0.020	0.618	0.766	0.966	0.714
impossible	0.655	0.663	0.014	0.872	0.237	0.561	0.679	0.870	0.844
chainlink	0.067	0.842	1	1	1	0.391	0.112	1	1
atom	0.293	0.201	1	1	0.294	0.579	0.040	1	0.969
mean (std)	0.487 (0.321)	0.544 (0.342)	0.596 (0.458)	0.708 (0.307)	0.569 (0.387)	0.644 (0.260)	0.635 (0.414)	0.968 (0.038)	0.930 (0.093)

The highest value for each data set is highlighted in each row. The mean value and standard deviation are shown in the last row. Same in the next table.

TABLE III
ARI OF THE TEN BENCHMARK PROBLEMS FROM THE EIGHT ALGORITHMS

Data set	K-Means++	GMM	SingleLinkage	HDBSCAN	MST	BP	Spectral	Gauging- δ	Gauging- δ (proximity)
3_blobs	0.99	0.99	0.99	0.955	0.57	0.965	0.99	0.985	0.985
3-spiral	-0.006	-0.006	1	0.05	1	0.172	1	0.973	1
Jain	0.324	-0.004	0.256	-0.108	0.904	0.23	1	1	1
compound	0.532	0.597	0.742	0.768	0.783	0.632	0.529	0.944	0.944
flame	0.431	0.29	0.013	0.324	0.013	0.955	0.013	0.988	0.988
aggregation	0.767	0.791	0.804	0.802	0.809	0.993	0.987	0.987	0.804
pathbased	0.464	0.437	0.001	0.788	0.001	0.564	0.683	0.98	0.598
impossible	0.585	0.531	0.002	0.831	0.057	0.157	0.602	0.841	0.814
chainlink	0.09	0.91	1	1	1	0.143	0.15	1	1
atom	0.182	0.081	1	1	0.379	0.574	0.002	1	0.988
mean (std)	0.436 (0.306)	0.462 (0.369)	0.581 (0.456)	0.641 (0.404)	0.552 (0.410)	0.539 (0.350)	0.596 (0.413)	0.970 (0.048)	0.912 (0.134)

TABLE IV
OVERVIEW OF THE MEAN AND STANDARD DEVIATION OF NMI W.R.T WELL-SEPARATED, NOT WELL-SEPARATED AND ALL THE DATA SETS OVER THE EIGHT METHODS WITH GROUND AVERAGE

	K-Means++	GMM	SingleLinkage	HDBSCAN	MST	BP	Spectral	Gauging- δ	Gauging- δ (proximity)
Well-separated (53)	0.598 (0.360)	0.683 (0.351)	0.905 (0.250)	0.885 (0.191)	0.701 (0.398)	0.687 (0.247)	0.592 (0.369)	0.754 (0.298)	0.941 (0.070)
Not well-separated (52)	0.673 (0.339)	0.694 (0.344)	0.227 (0.349)	0.649 (0.240)	0.164 (0.309)	0.678 (0.269)	0.662 (0.354)	0.541 (0.413)	0.511 (0.408)
All (105)	0.635 (0.350)	0.688 (0.346)	0.569 (0.455)	0.768 (0.246)	0.435 (0.446)	0.682 (0.257)	0.627 (0.362)	0.648 (0.374)	0.728 (0.361)

The sample size of each case is included, same below.

TABLE V
OVERVIEW OF THE MEAN AND STANDARD DEVIATION OF ARI W.R.T WELL-SEPARATED, NOT WELL-SEPARATED AND ALL THE DATA SETS OVER THE EIGHT METHODS WITH GROUND AVERAGE

	K-Means++	GMM	SingleLinkage	HDBSCAN	MST	BP	Spectral	Gauging- δ	Gauging- δ (proximity)
Well-separated (53)	0.550 (0.373)	0.636 (0.375)	0.892 (0.256)	0.861 (0.254)	0.658 (0.418)	0.572 (0.326)	0.551 (0.381)	0.666 (0.430)	0.942 (0.079)
Not well-separated (52)	0.650 (0.352)	0.664 (0.355)	0.159 (0.283)	0.599 (0.294)	0.116 (0.257)	0.595 (0.354)	0.630 (0.371)	0.491 (0.444)	0.480 (0.418)
All (105)	0.599 (0.364)	0.650 (0.364)	0.529 (0.456)	0.731 (0.303)	0.389 (0.440)	0.584 (0.338)	0.590 (0.377)	0.579 (0.444)	0.713 (0.377)

well-separated data sets. For data with more complex structures, Gauging- δ with continuity can improve performance, though further adjustments may be needed to prevent under-merging.

3) *Comparison With Parametric and Non-Parametric Algorithms*: The clustering results obtained from the other seven algorithms, together with results from both Gauging- δ and Gauging- δ (proximity) on the ten benchmark data sets are displayed in Table S1 (supplementary material). The NMI and ARI values corresponding to the benchmark problems in Fig. 6 are included in Tables II and III, respectively. Gauging- δ obtained the highest mean value and lowest standard deviation for both NMI and ARI among the eight algorithms. What's more, Gauging- δ got the highest NMI on six out of the ten data sets and the highest ARI on seven out of the ten data sets.

We can draw the following conclusions when we inspect the clustering results in Table S1 (supplementary material). In accordance with our expectation, the K-Means++ and GMM algorithms have difficulty identifying non-convex clusters ('spiral', 'Jain', 'chainlink', etc.). HDBSCAN sometimes does not

include border points in the clusters ('Jain', 'flame', 'aggregation'). BP often exhibits over-segmentation issues across various datasets and performs poorly with non-convex types, for example, clustering different spirals together in 'spiral'. MST struggles to partition close clusters (e.g., 'flame', 'aggregation', 'pathbased') using distance alone. The SingleLinkage hierarchical clustering algorithm shares similar drawbacks with MST in distinguishing the close clusters. Spectral clustering is good at dealing with non-convex data sets; however, it is not able to separate the large area touching in the 'flame' and overcut the 'impossible' and 'chainlink'.

The average value of the NMI indices from the eight algorithms on the total 105 data sets are shown in Table VI, with three averaging methods used. The ground average is obtained by simply averaging all the values. The categorized average is calculated by two steps. To eliminate the influence of the bias brought by the repeated similar data sets on the result, the indices of the data sets in the same series are averaged first, for example, 'sizes1', 'sizes2', 'sizes3', 'sizes4' and 'sizes5'

TABLE VI
THE MEAN AND STANDARD DEVIATION OF THE NMI ON THE TOTAL 105 DATA SETS WITH THREE AVERAGING WAYS

	K-Means++	GMM	SingleLinkage	HDBSCAN	MST	BP	Spectral	Gauging- δ	Gauging- δ (proximity)
Ground Average (105)	0.635 (0.350)	0.688 (0.346)	0.569 (0.455)	0.768 (0.246)	0.435 (0.446)	0.682 (0.257)	0.627 (0.362)	0.648 (0.374)	0.728 (0.361)
Categorized Average (88)	0.677 (0.319)	0.730 (0.306)	0.631 (0.429)	0.789 (0.245)	0.495 (0.442)	0.705 (0.243)	0.666 (0.336)	0.694 (0.341)	0.773 (0.327)
Unbiased Average (98)	0.626 (0.357)	0.681 (0.354)	0.589 (0.449)	0.768 (0.251)	0.445 (0.444)	0.672 (0.263)	0.617 (0.369)	0.672 (0.367)	0.746 (0.351)

The integers next to the averaging methods indicate the sample size being averaged. The two best-performing algorithms in each row are highlighted.

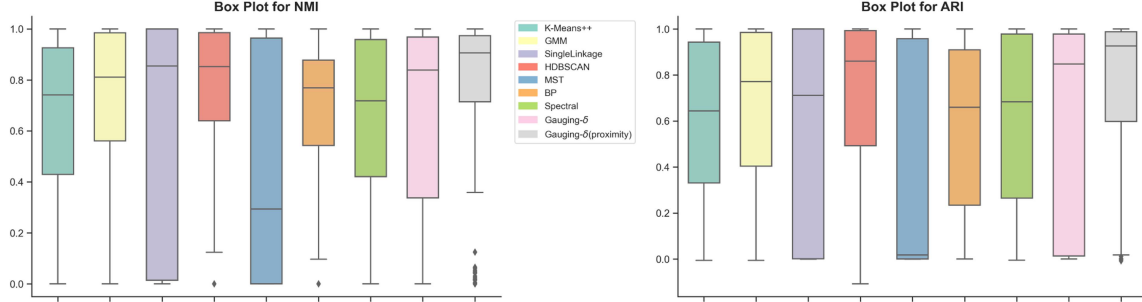


Fig. 7. Box plot of the NMI and ARI indices of the eight algorithms from the 105 data sets. The whiskers extend from the edges of the box to the farthest data point and extend no more than $1.5 * IQR$ ($IQR = Q3 - Q1$) from the edges of the box.

TABLE VII
THE CORRESPONDING STATISTICS OF THE NMI BOXPLOT IN FIG. 7

	K-Means++	GMM	SingleLinkage	HDBSCAN	MST	BP	Spectral	Gauging- δ	Gauging- δ (proximity)
Q1	0.429	0.561	0.014	0.64	0	0.543	0.421	0.337	0.714
Median (Q2)	0.741	0.811	0.855	0.852	0.294	0.769	0.718	0.839	0.906
Q3	0.926	0.985	1	0.986	0.964	0.878	0.959	0.968	0.974
1.5*IQR	0.746	0.636	1.479	0.519	1.446	0.502	0.807	0.946	0.39

The highest Q2 value and smallest 1.5*IQR among the algorithms are highlighted in bold.

TABLE VIII
THE CORRESPONDING STATISTICS OF THE ARI BOXPLOT IN FIG. 7

	K-Means++	GMM	SingleLinkage	HDBSCAN	MST	BP	Spectral	Gauging- δ	Gauging- δ (proximity)
Q1	0.331	0.404	0.001	0.493	0	0.234	0.265	0.013	0.598
Median (Q2)	0.644	0.771	0.712	0.86	0.018	0.66	0.683	0.848	0.926
Q3	0.943	0.985	1	0.993	0.958	0.909	0.978	0.978	0.988
1.5*IQR	0.918	0.871	1.498	0.75	1.437	1.013	1.07	1.448	0.585

The highest Q2 value and smallest 1.5*IQR among the algorithms are highlighted in bold.

are aggregated to ‘sizes’ and used as one sample in the final calculation. However, although some data sets in the same series are similar, an algorithm can produce very different results on them. For example, the GMM algorithm obtained NMI of 0.001, 0.011 and 1 on ‘long1’, ‘long2’ and ‘long3’, respectively. In this sense, the three similar data sets are not similar for GMM. Therefore, only the series of data sets that have similar output indices from every algorithm are averaged in unbiased average. The details of the averaging can be found in Table S7 and S8 (supplementary material). HDBSCAN got the highest NMI and Gauging- δ (proximity) ranks second under the three averaging methods.

We further compare the variability of the indices obtained from different methods by box plot, as shown in Fig. 7, Tables VII and VIII. Gauging- δ (proximity) exhibits the highest median values and the smallest IQR for both NMI and ARI, indicating that it achieves superior overall performance with lower variability in its results.

Therefore, our proposed algorithm, Gauging- δ (proximity), demonstrated consistently robust performance across diverse types of data sets, whereas other algorithms exhibited noticeable preferences for specific characteristics of data structure.

4) *Experiments Results on Real-World Data Sets:* In this section, we compare the proposed algorithm’s and other algorithms’

TABLE IX
NMI AND ARI OF THE ALGORITHMS ON THE USPS DATA SET UNDER DIFFERENT DISTANCE METRICS AND FEATURES

	Setting 1		Setting 2		Setting 3	
	NMI	ARI	NMI	ARI	NMI	ARI
K-Means++	0.598	0.497	-	-	0.631 $\uparrow$	0.596 $\uparrow$
GMM	0.591	0.475	-	-	0.625 $\uparrow$	0.586 $\uparrow$
SingleLinkage	0.009	0	0.009	0	0.011 $\uparrow$	0.001 $\uparrow$
HDBSCAN	0.278	0.087	0	0	0.303 $\uparrow$	0.054
MST	0.49	0.203	0	0	0.587 $\uparrow$	0.475 $\uparrow$
BP	0.086	0.011	-	-	0.647 $\uparrow$	0.583 $\uparrow$
Spectral	0.357	0.132	0.694 $\uparrow$	0.561 $\uparrow$	0.679 $\uparrow$	0.603 $\uparrow$
Gauging- δ	0.41	0.26	0.619 $\uparrow$	0.397 $\uparrow$	0.585 $\uparrow$	0.314 $\uparrow$

In setting 2, K-Means++, GMM, and BP cannot take the dissimilarity matrix as input. The upper arrow near a number indicates the index increases compared to the baseline setting (setting 1).

performance on the real-world high-dimensional data sets and explore the influence of data features and distance metrics to the clustering results. Table IX displays the NMI and ARI results for the proposed algorithm alongside various comparison algorithms under three settings, original data features with pixel-wise euclidean distance (setting 1), a dissimilarity measure designed in this paper (setting 2), and CNN encoded data features with euclidean distance (setting 3).

In clustering, accurately determining the distance between clusters is crucial for effective merging decisions. In the USPS application example, two clusters will not merge unless the average distance between all points in both clusters is sufficiently small, rather than just the nearest points being close. To address this, we use the average distance between all pairs of points from the two clusters to determine their proximity during the clustering process in Gauging- δ .

In the baseline setting (setting 1), the Gauging- δ achieved slightly lower performance in two indices compared to K-Means++, GMM, MST, and spectral clustering, but outperformed HDBSCAN, BP, and SingleLinkage. The SingleLinkage and BP methods struggled significantly, with their indices nearing zero, indicating poor clustering accuracy.

To explore the impact of different distance metrics on clustering performance, we conducted experiments using euclidean

(setting 1) and the designed dissimilarity measure (setting 2), respectively. The dissimilarity measure, designed specifically in this paper, calculates the distance between two images by summing the shortest distances between each pixel and its closest counterpart in the other image. This metric is more effective at capturing image similarities. As shown in Table IX, the dissimilarity measure significantly enhances the Gauging- δ 's performance, while SingleLinkage, HDBSCAN, and MST would cluster all the data together, resulting in an index at zero.

As shown in Table IX, all the algorithm's performance has improved when using the CNN-transformed features (setting 3) compared to the original data (setting 1). The CNN was implemented with the following structure: it begins with an input layer for 16x16 grayscale images, followed by two convolutional layers with 32 and 64 filters, both utilizing ReLU activation. Each convolutional layer is followed by a max pooling layer with a (2, 2) pool size. The resulting feature maps are then flattened and passed through a dense layer consisting of 128 units with ReLU activation. The final output layer has 10 units with softmax activation. After training, an intermediate model was used to extract 128-dimensional embeddings from the dense layer, which were applied to convert the test set images into high-level feature vectors.

Therefore, the effective application of our proposed algorithm depends on employing data features and similarity measures suited to each domain's specific challenges.

IV. DISCUSSIONS

We present Gauging- δ and its proximity-based variant, Gauging- δ (proximity), for clustering data sets with diverse shapes, densities, and distributions. Gauging- δ (proximity) proves a near-perfect performance for well-separated data sets and demonstrates the ability to handle many not well-separated data sets. Compared to other algorithms that displayed clear preferences for specific data structures, Gauging- δ (proximity) demonstrates consistently robust performance across a wide range of data sets. For more complex data structures, the Gauging- δ with continuity shows improved clustering performance, though it introduced certain under-merging for well-separated clusters. This investigation into the algorithm's structure offers valuable insights for method selection and mergeability function refinement based on the characteristics of domain-specific data sets. The ultimate performance depends on tailoring data features and similarity measures to meet the unique challenges posed by each domain. Future work will refine the continuity function to further improve the performance on not well-separated data sets.

In application, the effectiveness of the algorithm we propose relies on developing better metrics for mergeability tailored to the specific application domain. As Luxburg [41] pointed out, it is not feasible to create a one-size-fits-all clustering algorithm without considering the particularities of the application context. The success of a clustering algorithm is tied to the application domain—it requires the right choice of data features and distance or similarity metrics which in turn requires a thorough understanding of the domain and the specific challenges it presents.

The proposed algorithm's non-parametric design does not rely on prior knowledge of the data, unlike many other clustering algorithms. However, it does require a well-defined similarity measure, which can either be determined through domain investigation or learned from the data [42].

In conclusion, the algorithm can be improved by selecting appropriate features, designing representative similarity metrics, and a mergeability function that is appropriate for the specific application.

V. CONCLUSION

The paper presents a novel non-parametric adaptive clustering algorithm that is designed to handle various shape clusters and various spatial cluster relations. The algorithm employs an adaptive mergeability function to iteratively merge clusters based on their statistical properties and environment information. The clustering process can stop automatically when no pair of clusters can be merged without specifying the number of clusters.

The algorithm is shown to outperform several representative clustering methods on a range of benchmark datasets, especially on well-separated data sets. It is noteworthy that the test was done on over one hundred datasets for the first time, providing a more confident demonstration of the proposed algorithm's advantages.

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REFERENCES

- [1] R. Xu and D. Wunsch, "Survey of clustering algorithms," *IEEE Trans. Neural Netw.*, vol. 16, no. 3, pp. 645–678, May 2005.
- [2] A. K. Jain, M. N. Murty, and P. J. Flynn, "Data clustering: A review," *ACM Comput. Surv.*, vol. 31, no. 3, pp. 264–323, 1999.
- [3] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Math. Statist. Probability*, 1967, pp. 281–298.
- [4] A. E. Ezugwu et al., "A comprehensive survey of clustering algorithms: State-of-the-art machine learning applications, taxonomy, challenges, and future research prospects," *Eng. Appl. Artif. Intell.*, vol. 110, Apr. 2022, Art. no. 104743.
- [5] D. Cheng, J. Huang, S. Zhang, S. Xia, G. Wang, and J. Xie, "K-means clustering with natural density peaks for discovering arbitrary-shaped clusters," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 11077–11090, Aug. 2024.
- [6] A. K. Jain, "Data clustering: 50 years beyond K-means," *Pattern Recognit. Lett.*, vol. 31, no. 8, pp. 651–666, 2010.
- [7] D. Cheng, Q. Zhu, J. Huang, Q. Wu, and L. Yang, "A novel cluster validity index based on local cores," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 4, pp. 985–999, Apr. 2019.
- [8] D. Cheng, Q. Zhu, J. Huang, Q. Wu, and L. Yang, "Clustering with local density peaks-based minimum spanning tree," *IEEE Trans. Knowl. Data Eng.*, vol. 33, no. 2, pp. 374–387, Feb. 2021.
- [9] M. Ronen, S. E. Finder, and O. Freifeld, "DeepDPM: Deep clustering with an unknown number of clusters," in *Proc. 2022 IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 9851–9860.
- [10] S. Sarfraz, V. Sharma, and R. Stiefelhofen, "Efficient parameter-free clustering using first neighbor relations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 8934–8943.

- [11] B. Mukhoty, R. Gupta, and L. K. M. Kumar, "A parameter-free affinity based clustering," *Appl. Intell.*, vol. 50, pp. 4543–4556, Dec. 2020.
- [12] N. Leopold and O. Rose, "UNIC: A fast nonparametric clustering," *Pattern Recognit.*, vol. 100, Apr. 2020, Art. no. 107117.
- [13] A. U. Rehman and S. B. Belhaouari, "Divide well to merge better: A novel clustering algorithm," *Pattern Recognit.*, vol. 122, 2022, Art. no. 108305.
- [14] H. Averbuch-Elor, N. Bar, and D. Cohen-Or, "Border-peeling clustering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 7, pp. 1791–1797, Jul. 2020.
- [15] M. Du, R. Wang, R. Ji, X. Wang, and Y. Dong, "ROBP a robust border-peeling clustering using Cauchy kernel," *Inf. Sci.*, vol. 571, pp. 375–400, Sep. 2021.
- [16] Z. Kang et al., "Structured graph learning for clustering and semi-supervised classification," *Pattern Recognit.*, vol. 110, Feb. 2021, Art. no. 107627.
- [17] Z. Kang, Z. Lin, X. Zhu, and W. Xu, "Structured graph learning for scalable subspace clustering: From single view to multiview," *IEEE Trans. Cybern.*, vol. 52, no. 9, pp. 8976–8986, Sep. 2022.
- [18] D. Cheng, J. Huang, S. Zhang, X. Zhang, and X. Luo, "A novel approximate spectral clustering algorithm with dense cores and density peaks," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 52, no. 4, pp. 2348–2360, Apr. 2022.
- [19] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowl. Discov. Data Mining*, Portland, Oregon, 1996, pp. 226–231.
- [20] A. Rodriguez and A. Laio, "Clustering by fast search and find of density peaks," *Science*, vol. 344, pp. 1492–1496, Jun. 2014.
- [21] D. Cheng, Y. Li, S. Xia, G. Wang, J. Huang, and S. Zhang, "A fast granular-ball-based density peaks clustering algorithm for large-scale data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 17202–17215, Dec. 2024.
- [22] R. Sibson, "SLINK: An optimally efficient algorithm for the single-link cluster method," *Comput. J.*, vol. 16, pp. 30–34, Jan. 1973.
- [23] F. Ros and S. Guillaume, "A hierarchical clustering algorithm and an improvement of the single linkage criterion to deal with noise," *Expert Syst. Appl.*, vol. 128, pp. 96–108, Aug. 2019.
- [24] E. K. Tokuda, C. H. Comin, and L. D. F. Costa, "Revisiting agglomerative clustering," *Physica A, Stat. Mech. Appl.*, vol. 585, Jan. 2022, Art. no. 126433.
- [25] Y. Zeng, T. A. Nguyen, B. Yan, and S. Li, "A distance-based parameter free algorithm for curve reconstruction," *Comput.-Aided Des.*, vol. 40, pp. 210–222, Feb. 2008.
- [26] T. A. Nguyen and Y. Zeng, "VICUR: A human-vision-based algorithm for curve reconstruction," *Robot. Comput.-Integr. Manuf.*, vol. 24, pp. 824–834, Dec. 2008.
- [27] J. Yang and C.-T. Lin, "Autonomous clustering by fast find of mass and distance peaks," *IEEE Trans. Pattern Anal. Mach. Intell.*, pp. 1–14, 2025, doi: [10.1109/TPAMI.2025.3535743](https://doi.org/10.1109/TPAMI.2025.3535743).
- [28] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [29] H. Chang and D.-Y. Yeung, "Robust path-based spectral clustering," *Pattern Recognit.*, vol. 41, no. 1, pp. 191–203, 2008.
- [30] A. Fred and A. Jain, "Data clustering using evidence accumulation," in *Proc. 2002 Int. Conf. Pattern Recognit.*, 2002, pp. 276–280.
- [31] C. Zahn, "Graph-theoretical methods for detecting and describing gestalt clusters," *IEEE Trans. Comput.*, vol. C-20, no. 1, pp. 68–86, Jan. 1971.
- [32] L. Fu and E. Medico, "FLAME, a novel fuzzy clustering method for the analysis of DNA microarray data," *BMC Bioinf.*, vol. 8, no. 1, 2007, Art. no. 3.
- [33] A. Gionis, H. Mannila, and P. Tsaparas, "Clustering aggregation," *ACM Trans. Knowl. Discov. Data*, vol. 1, no. 1, pp. 4–es, 2007.
- [34] M. C. Thrun and A. Ullsch, "Clustering benchmark datasets exploiting the fundamental clustering problems," *Data Brief*, vol. 30, 2020, Art. no. 105501.
- [35] L. H. D. S. Fernandes, A. C. Lorena, and K. Smith-Miles, "Towards understanding clustering problems and algorithms: An instance space analysis," *Algorithms*, vol. 14, Mar. 2021, Art. no. 95.
- [36] C. Zhong, D. Miao, and R. Wang, "A graph-theoretical clustering method based on two rounds of minimum spanning trees," *Pattern Recognit.*, vol. 43, pp. 752–766, Mar. 2010.
- [37] D. Arthur and S. Vassilvitskii, "k-means++: The advantages of careful seeding," in *Proc. 18th Annu. ACM-SIAM Symp. Discrete Algorithms*, 2007, pp. 1027–1035.
- [38] C. Fraley and A. Raftery, "Model-based clustering, discriminant analysis, and density estimation," *J. Amer. Stat. Assoc.*, vol. 97, no. 458, pp. 611–631, 2002.
- [39] R. J. G. B. Campello, D. Moulavi, and J. Sander, "Density-based clustering based on hierarchical density estimates," in *Proc. Adv. Knowl. Discov. Data Mining*, Berlin, Heidelberg, Springer, 2013, pp. 160–172.
- [40] S. X. Yu and J. Shi, "Multiclass spectral clustering," in *Proc. IEEE 9th Int. Conf. Comput. Vis.*, 2003, pp. 313–319.
- [41] U. V. Luxburg, R. C. Williamson, and I. Guyon, "Clustering: Science or art?," in *Proc. ICML Workshop Unsupervised Transfer Learn.*, 2012, pp. 65–79.
- [42] Z. Kang, H. Pan, S. C. H. Hoi, and Z. Xu, "Robust graph learning from noisy data," *IEEE Trans. Cybern.*, vol. 50, no. 5, pp. 1833–1843, May 2020.



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